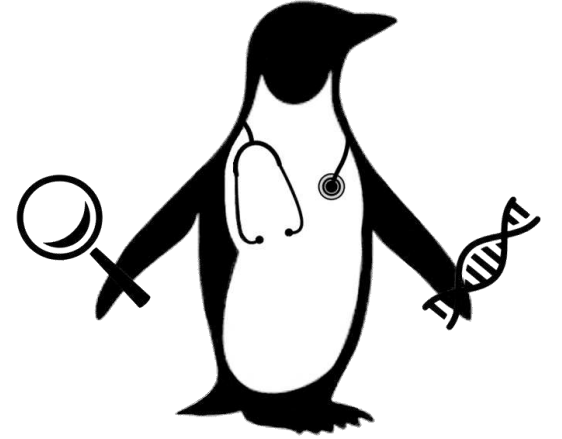


PENGQUIN:

PErsoNal Genome QUery IN healthcare and clinical practice



Elias Crum

FWO Strategic Basic PhD Fellowship Interview
September 5th, 2024
Pannel: SBWT5B

Personal Genome Data Usage?

My Journey

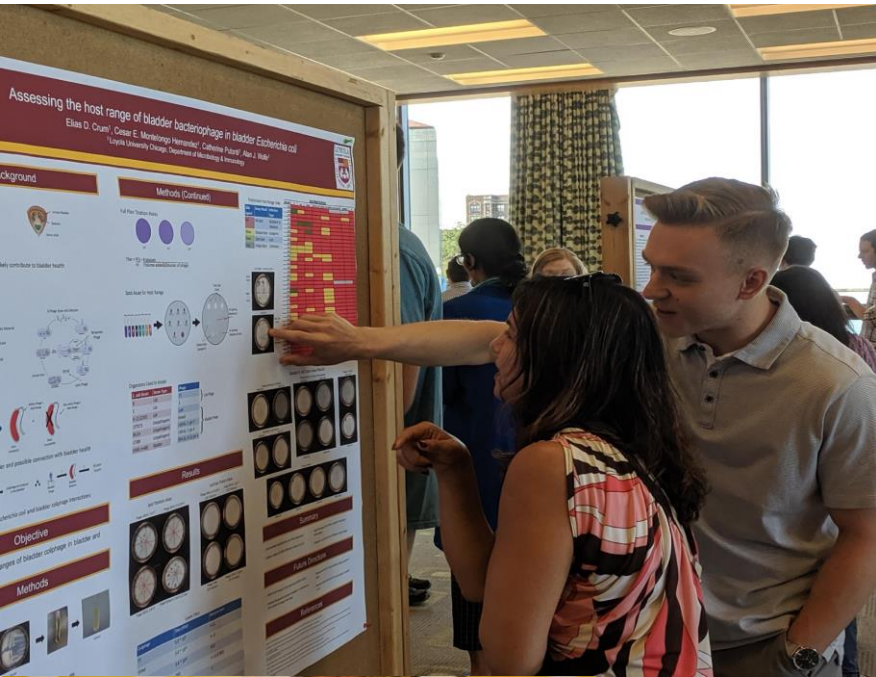


Medicine + Bioinformatics



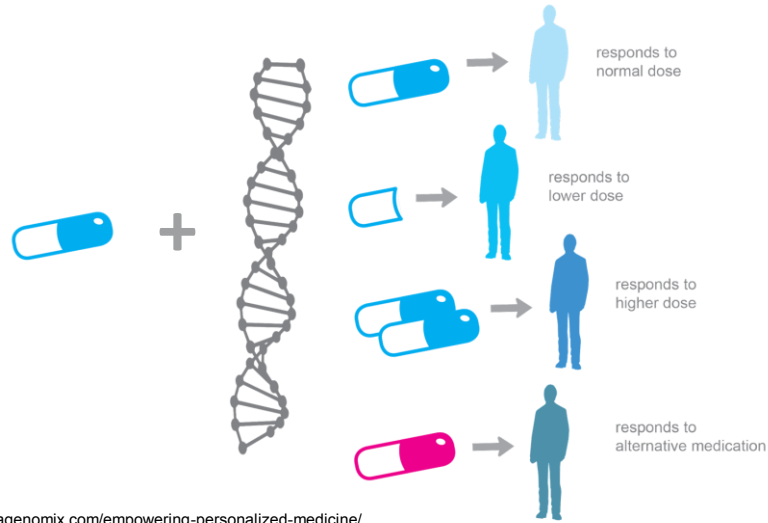
Genomics

How can we increase **genomics** use in **medicine**?



Genomics + Healthcare

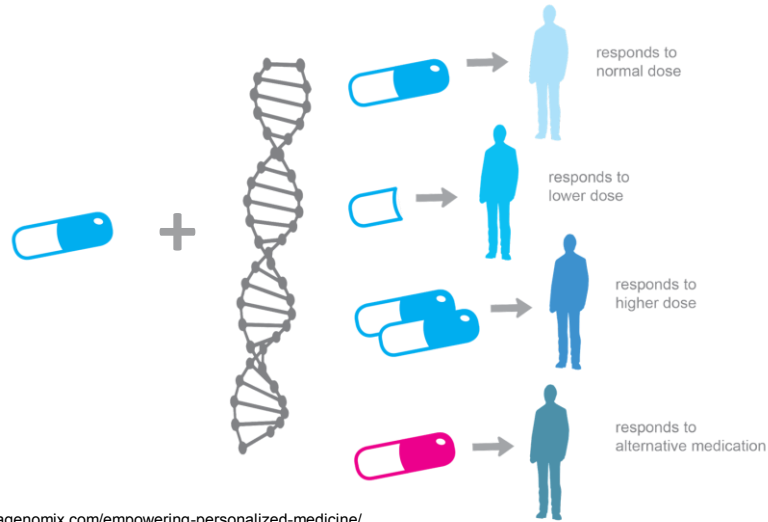
A *Example: Pharmacogenomics*



<https://alphagenomix.com/empowering-personalized-medicine/>

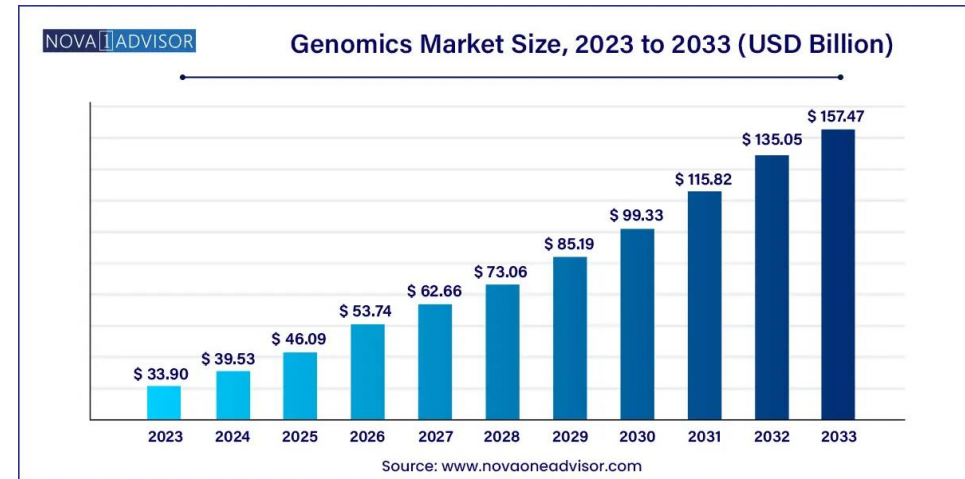
Genomics + Healthcare

A Example: Pharmacogenomics



<https://alphagenomix.com/empowering-personalized-medicine/>

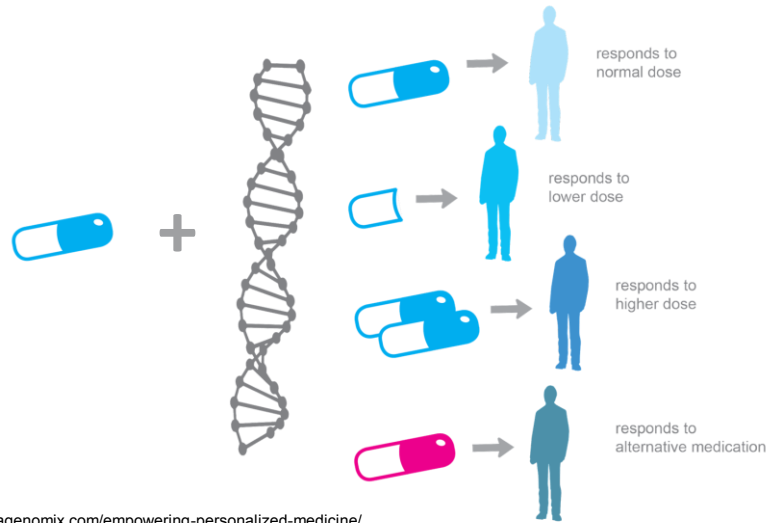
B Genomics market is growing RAPIDLY



<https://www.novaoneadvisor.com/report/genomics-market>

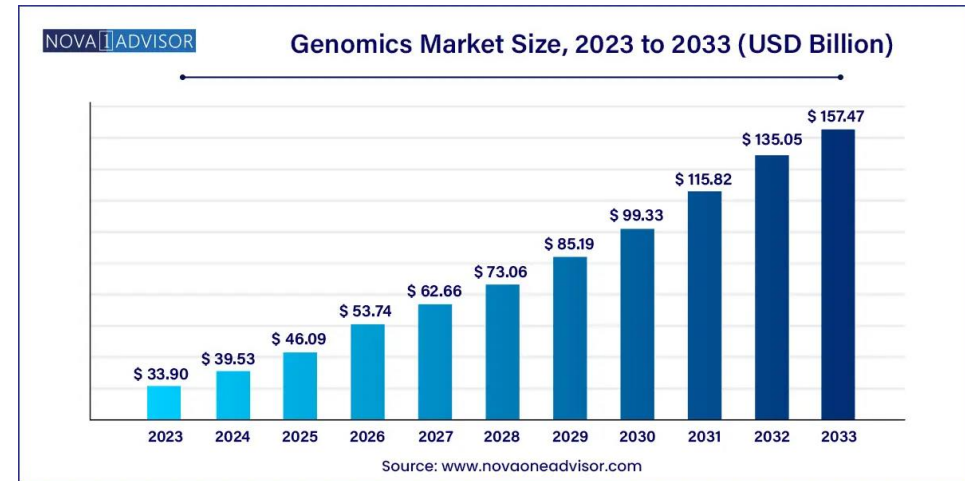
Genomics + Healthcare

A Example: Pharmacogenomics



<https://alphagenomix.com/empowering-personalized-medicine/>

B Genomics market is growing RAPIDLY



<https://www.novaoneadvisor.com/report/genomics-market>

C

Institution-centric

Data Storage



Infrastructure



?

Data Usage



**Low Efficiency
+
High Costs**

Problem:

Personal Genome Sequence Data Challenges

Size considerations:

*Variant Call Format (**VCF**) file = ~200 MB*

Represents ~3 million mutations

Formatted as a flat (.txt) file

Other considerations:



Governance



Privacy



Formatting



Storage

PENGQUIN Goals

Improve genomic data **USAGE** and **SCALABILITY**



Improve genomic data store accessibility (while maintaining privacy)



Represent genomic data semantically as Linked Data



Query genomic data and linkages



Connect to existing initiatives / applications

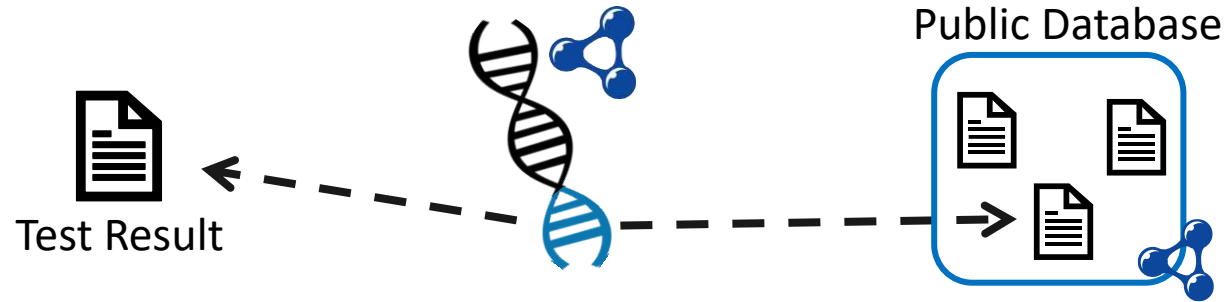
Proof-of-concept



Develop a framework and web application

Primary Research Challenges

WP1. Linking Genomic data to clinically-relevant data

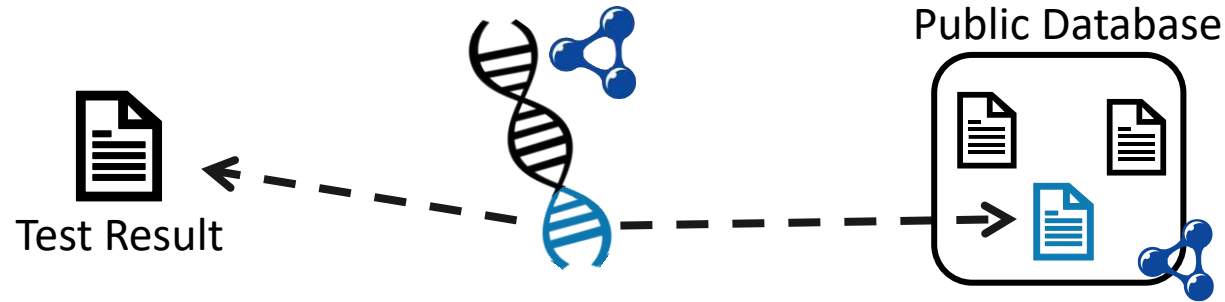


Genomic data to RDF?

Automation of Linkages?

Primary Research Challenges

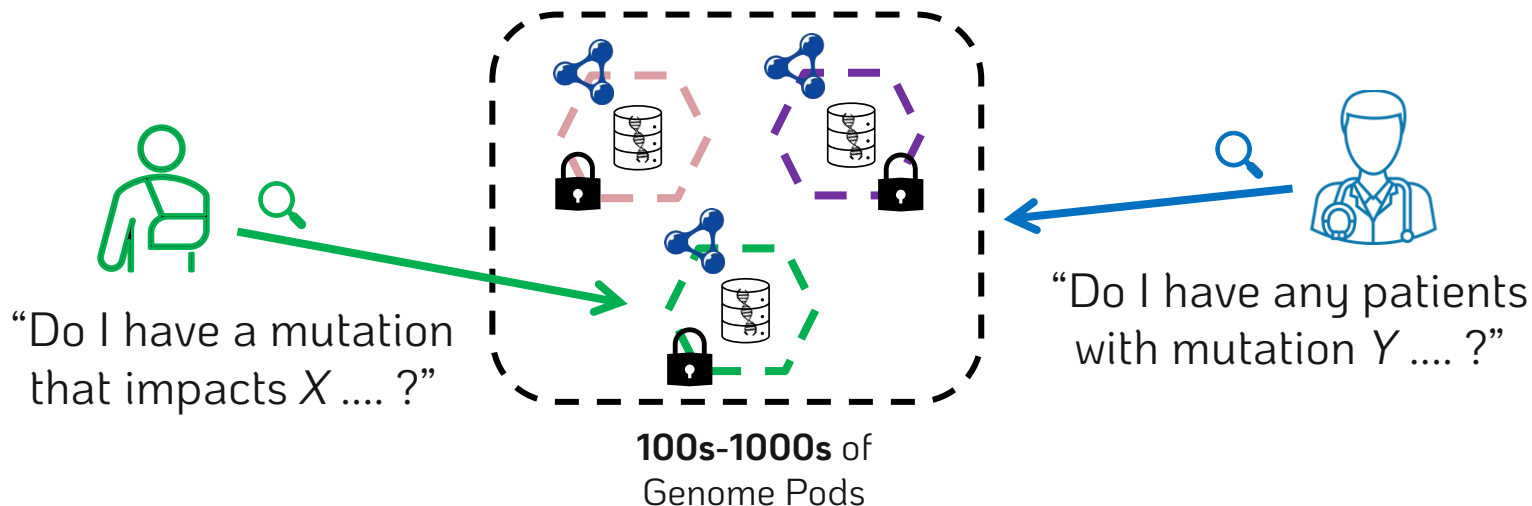
WP1. Linking Genomic data to clinically-relevant data



Genomic data to RDF?

Automation of Linkages?

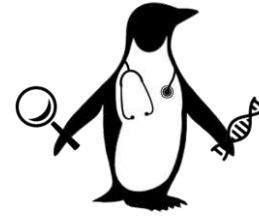
WP3. Efficiently finding data through querying



MANY, LARGE sources?



Genomic Data



Clinical Care



Semantic, Linked Data



Query Capabilities



Framework + App

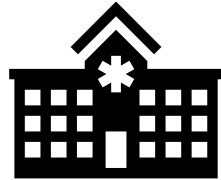
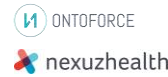
Supplemental Slides

Envisioned Product Evolution



PhD End

PhD
Deliverable



Short Term

Start-up / Product
development
+
Hospital implementation



Medical record



Medium Term

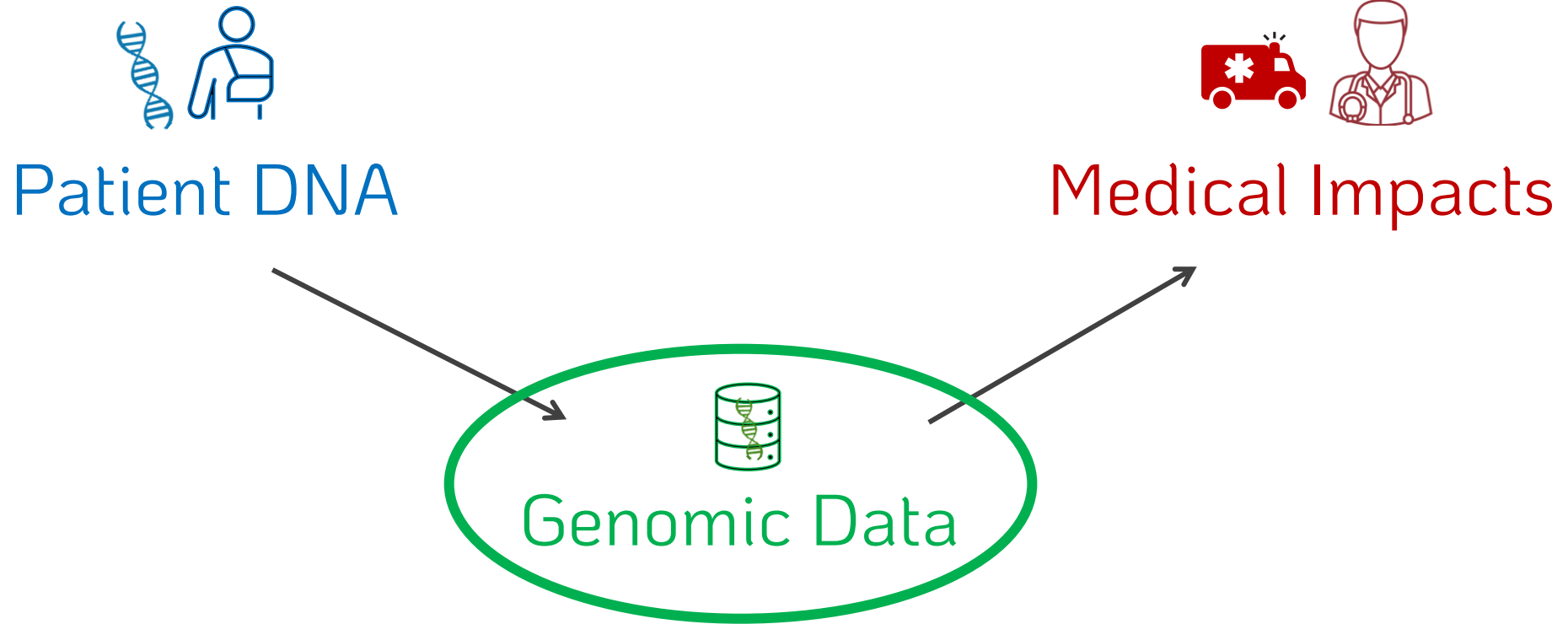
Integrate with EHR
+
Increase application
domains



Long Term

Increase 3rd party
application support
+
Scale to Europe and
beyond

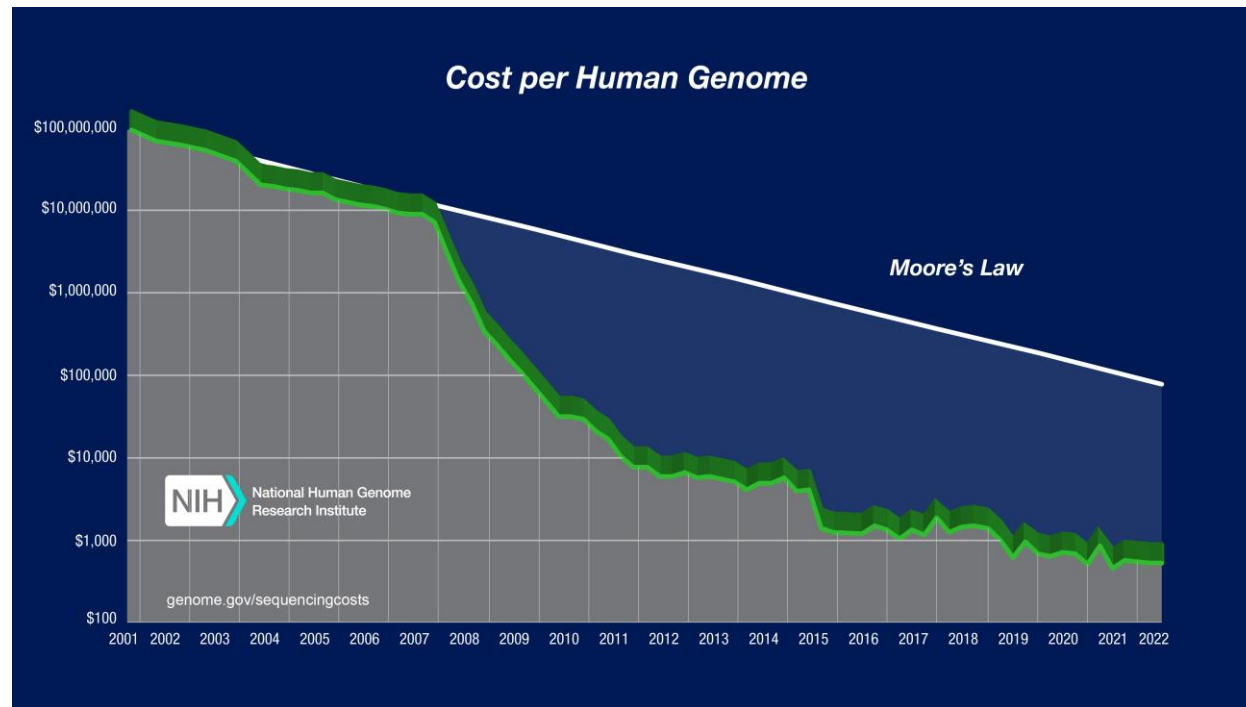
Why this Panel?



PERsoNal Genome QUery IN healthcare and clinical practice

Genome Sequencing in Health Care

Sequence Generation Costs

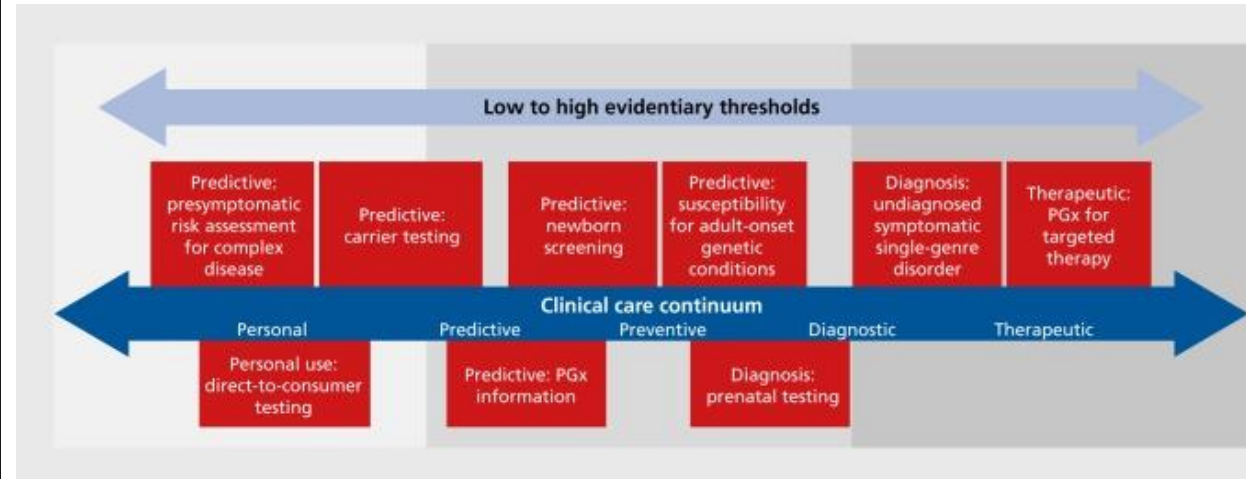


Wetterstrand KA. Accessed 12/04/24.

Currently getting close to \$100 per genome⁴

Per-base cost of sequencing has been dropping by half every ~5 months
Per-base cost of sequence data storage is dropping by half every ~14 months

Use-cases



10.31887/DCNS.2016.18.3/jkrier

Examples include:

- Medication prescription (pharmacogenomics)
- Genetic disease screening
- Cancer diagnosis & treatment

Personal Genome Sequence Data

Human genome data is big...

~3.2 BILLION base pairs ----> ~1 GB

Sequencing output file ----> ~200 GB

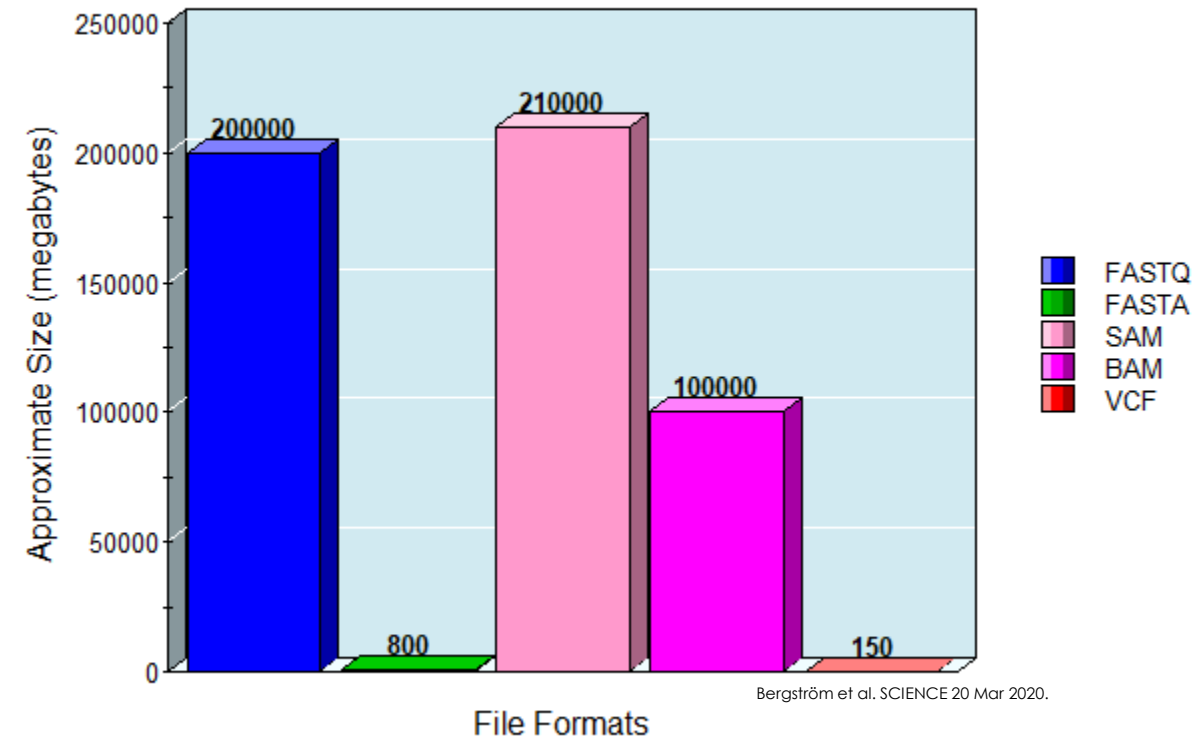
Variant Call Format (VCF) file ----> ~200 MB

- Individual humans exhibit ~3 million mutations
- VCF file **records only those differences**
- Flat file

VCF file semantic representation

- ~ >**27 MILLION** Triples when represented as RDF
- Ontology for conversion
- Header Dictionary Triples (for compression)

Sequencing File Sizes



Sequence Data Formats



Raw Sequencing Outputs

Contiguous DNA sequence

FASTQ reads aligned to a FASTA reference

Areas in SAM where sample and reference differ

```
##fileformat=VCFv4.3
##fileDate=20090805
##source=Imputation/ProgramV3.1
##reference=file:///seq/references/1000GenomesPilot-NCBI36.fasta
##contig=chr1,length=62435964,assembly=B36,mdb=1f26c8fa6e037f7949618ff66bb72da,species="Homo sapiens",taxonomy=xp
##phasing=format
##INFO=<ID=NS,Number=1,Type=Integer,Description="Number of Samples With Data">
##INFO=<ID=DP,Number=1,Type=Integer,Description="Total Depth">
##INFO=<ID=AF,Number=4,Type=Float,Description="Allele Frequency">
##INFO=<ID=AA,Number=1,Type=String,Description="Ancestral Allele">
##INFO=<ID=H2,Number=0,Type=Flag,Description="dSNP membership, build 129">
##INFO=<ID=H2,Number=0,Type=Flag,Description="HapMap2 membership">
##FILTER=<ID=gq10,Description="Quality below 10">
##FILTER=<ID=c50,Description="Less than 50% of samples have data">
##FORMAT=<ID=GT,Number=1,Type=String,Description="Genotype">
##FORMAT=<ID=GQ,Number=1,Type=Integer,Description="Genotype Quality">
##FORMAT=<ID=DP,Number=1,Type=Integer,Description="Read Depth">
##FORMAT=<ID=H2,Number=2,Type=Integer,Description="Haplotype Quality">
#CHROM POS ID REF ALT QUAL FILTER INFO FORMAT NAO0001 NAO0002 NAO0003
20 14370 rs6054257 G A 29 PASS NS=3;DP=14;AF=0.5;DB;H2 GT/GQ/DP/H2 0148:1:51,51 110:48:51,51 1/1:43:51,
20 17330 . G C 9 G10 PASS NS=3;DP=14;AF=0.017 GT/GQ/DP/H2 013:3:58,50 013:3:56:5,3 0/1:40:51,
20 1110696 rs6040355 G T 67 PASS NS=2;DP=10;AF=0.333,0.667;AA=T;DB GT/GQ/DP/H2 112:41:61,23,27 111:21:18,2 2/2:34,
20 1230237 . G T 47 PASS NS=3;DP=13;AA=T GT/GQ/DP/H2 0154:7:56,60 010:48:4:51,51 0/0:61:2
20 1248467 microsat GTC G,GTCT 50 PASS NS=3;DP=9;AA=T GT/GQ/DP/H2 01:136:4 0/2:17:2 1/1:40:3
```

Diagram illustrating the generation of a sequence from a label and base:

- Label** (e.g., @FORJUSP02AJWD1)
- Sequence** (e.g., CGTCGAATT CATT TAAAGTTT TAACCTT GCGGCCGT ACTCCCAGGCGGT)
- Q scores (as ASCII chars)** (e.g., :99@:::??@:::FFAAAAACCAA:::BB@@?A?)
- Base=T, Q=': '=25**

The sequence is generated by combining the label and the base, resulting in the final sequence: @FORJUSP02AJWD1CGTCGAATT CATT TAAAGTTT TAACCTT GCGGCCGT ACTCCCAGGCGGT + AAAAAAAAAA:::99@:::??@:::FFAAAAACCAA:::BB@@?A?

© Elias Crum 2026

Sequencing Storage

The human genome

=

~3.2 billion base pairs



- 3 billion letters -- ~700 megabytes

Representative of a **single strand** of the whole human genome sequence

E.g. : ...ATGCCGTAAACAGATGTCA ...

Format: *TXT* file

- Raw human whole genome sequence -- ~200 gigabytes

Whole genome sequences typically have **>=30x coverage at each base position**. (i.e. each base of the genome is represented in the sequence file a ≥ 30 times)

Format: *FASTQ* file

- Aligned human whole genome sequence -- ~210 gigabytes // ~100 gigabytes // ~15 gigabytes

Whole genome sequence **is mapped to a reference human genome** and can be stored in various formats. Recent advances have allowed for improved compression.

Format: *SAM* file // *BAM* file // *CRAM* file

- Human genome variant file (list of mutations) -- ~125 megabytes

Individual humans only exhibit **~0.1% variation** comparatively (~3 million mutations). Variant file **records those differences** in one individual's genome compared to the “normal” reference genome.

Format: *VCF* file

Variant Call Format (VCF)

- tab delimited .txt file
- Rows represent individual mutations in a sequence (when compared to reference)
- Columns store information about that mutation

#CHROM	POS	ID	REF	ALT	QUAL	FILTER	INFO	FORMAT
--------	-----	----	-----	-----	------	--------	------	--------



```
##fileformat=VCFv4.1
##fileDate=20140930
##source=23andme2vcf.pl https://github.com/arrogantrobot/23
##reference=file:///23andme_v3_hg19_ref.txt.gz
##FORMAT=<ID=GT,Number=1,Type=String,Description="Genotype"
#CHROM POS ID REF ALT QUAL FILTER INFO FORMAT GEN
chr1 82154 rs4477212 a . . . . GT 0/0
chr1 752566 rs3094315 g A . . . . GT 1/1
chr1 752721 rs3131972 A G . . . . GT 1/1
chr1 798959 rs11240777 g . . . . GT 0/0
chr1 800007 rs6681049 T C . . . . GT 1/1
chr1 838555 rs4970383 c . . . . GT 0/0
chr1 846808 rs4475691 C . . . . GT 0/0
chr1 854250 rs7537756 A . . . . GT 0/0
chr1 861808 rs13302982 A G . . . . GT 1/1
chr1 873558 rs1110052 G T . . . . GT 1/1
chr1 882033 rs2272756 G A . . . . GT 0/1
chr1 888659 rs3748597 T C . . . . GT 1/1
chr1 891945 rs13303106 A G . . . . GT 0/1
```

Experimental VCF Data

Primary dataset from publicly available repositories (>100 genomes)

- Personal Genome Project (https://my.pgp-hms.org/public_genetic_data)
- NIST's Genome in a Bottle public dataset (<https://data.nist.gov/od/id/mds2-2336>)
- Illumina Platinum Genomes repository (<https://emea.illumina.com/platinumgenomes.html>)

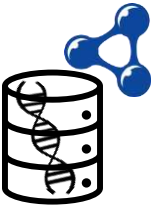
If more are needed (>1000 [research grade] genomes):

- 1000 genomes project (<https://www.internationalgenome.org/>)
- NCBI's ClinVar (<https://www.ncbi.nlm.nih.gov/clinvar/>)



- Does not solve ALL problems, but offers an environment where improvements are possible
 - ✓ Easier sharing (web-facilitated)
 - ✓ Granular privacy controls
 - ✓ Direct patient data access
 - ✓ Data representation flexibility (RDF)
 - ✓ Possibility of data-linking and querying

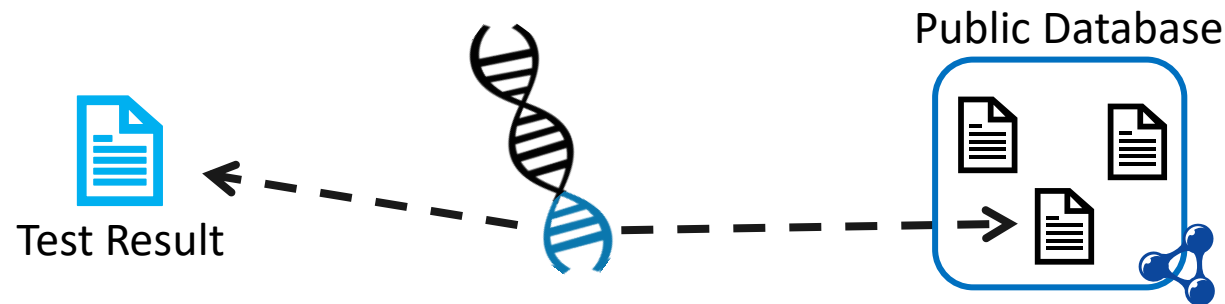
Genome Data in RDF



VCF genomic data can be represented as RDF.

 VCF to RDF Vocabularies exist $\longrightarrow$ SPHN Semantic Interoperability Framework⁴
+ FHIR HL7 Format⁵

VCF genomic data can be Linked to other data.



How to automate this?

What vocabularies could be used for the linkages?



Querying Genome Pods



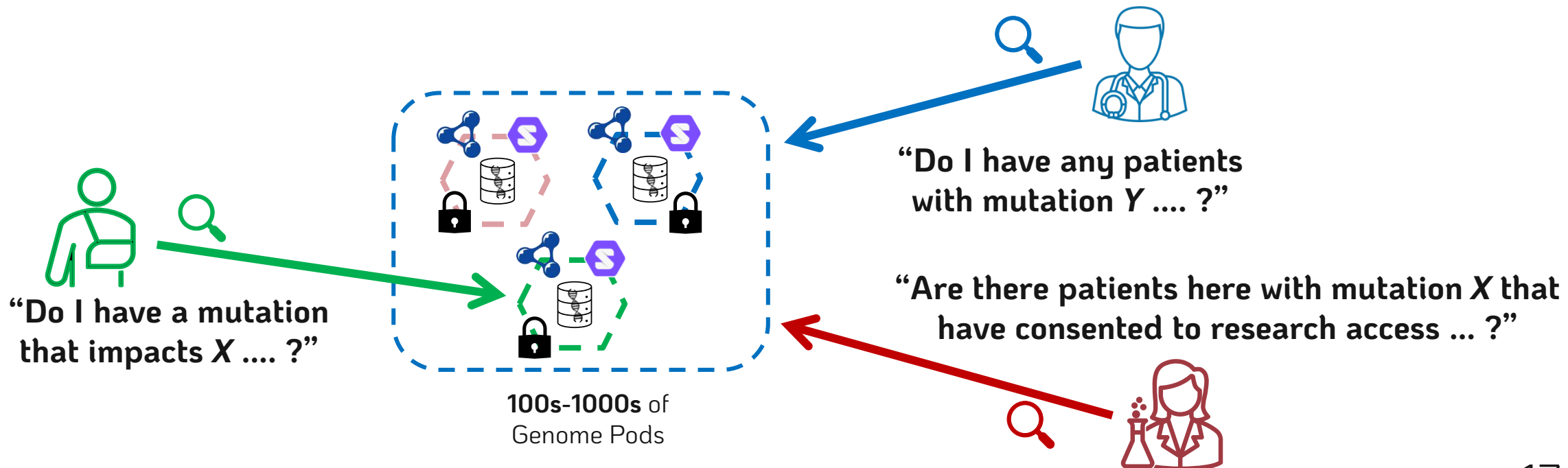
RDF genomic data pods can be queried.



Query Engine



Custom Comunica implementation/engine⁶



Querying Genome Pods (possibilities)



Single Pod Query

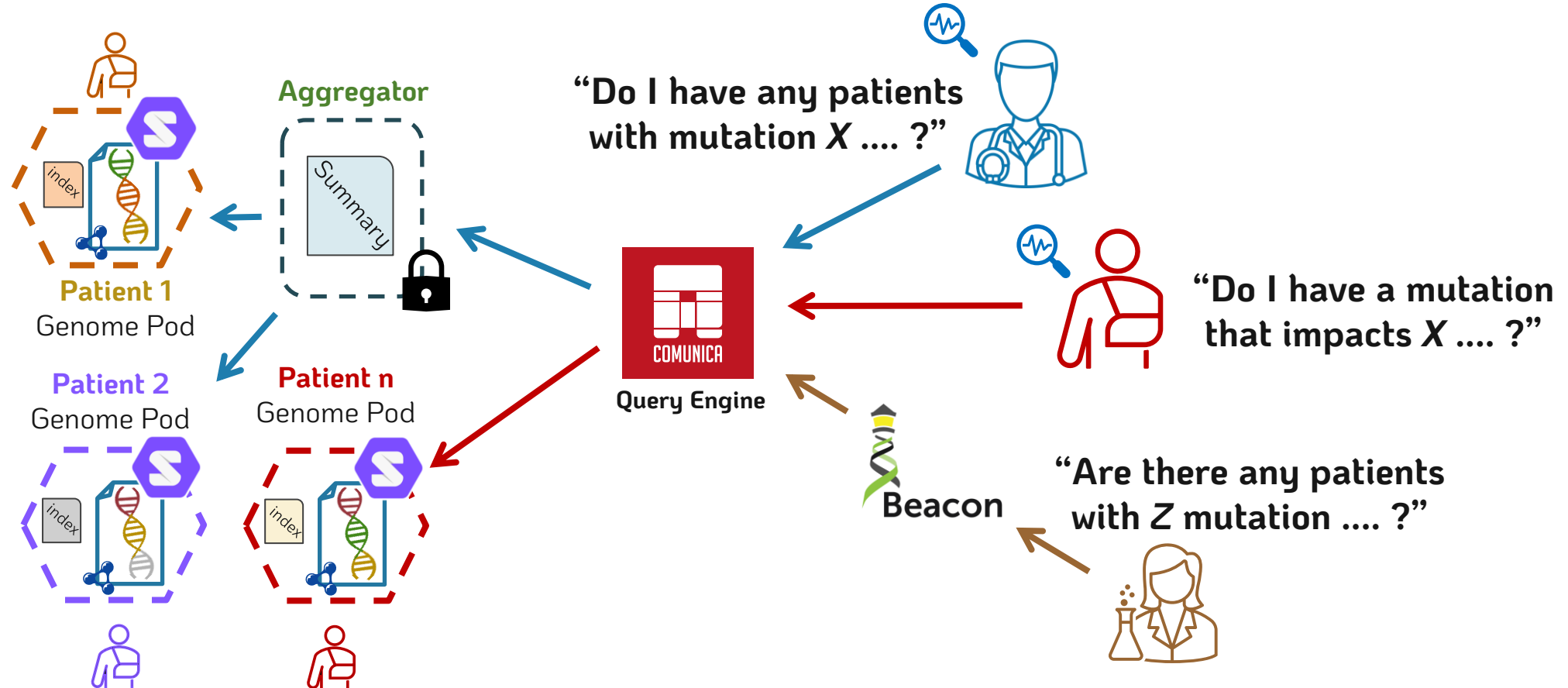
Utilizes an intra-pod index

Many Pod Query

First, utilizes an extra-pod aggregator
Then, utilizes the intra-pod indexes

Beacon API for Researcher Queries

Query is translated from Beacon API to Comunica
Then, the **Many Pod** query approach is followed



Implementation Specifics



Solid pod instances → Community Solid Server¹

Genomic Data → Publicly available *VCF* data²

RDF Conversion → Aim to use SPHN RDF vocabulary⁴

Data-linking → Actively exploring ontologies / methodologies

Data Querying → Custom instance of Comunica⁶